

Session-06: Poisson Distribution

1. The random variable X has a Poisson distribution. If $P(X = 1) = 0.01487$, $P(X = 2) = 0.04461$, then find $P(X = 3)$.

Ans: 0.08922

2. Wireless sets are manufactured with 25 soldered joints each. On the average, 1 joint in 500 is defective. How many sets can be expected to be free from defective joints in a consignment of 10000 sets?

Ans: 9512

3. Suppose the number of telephone calls on an operator received from 9.00 to 9.25 follow a Poisson distribution with mean 3. Find the probability that
- The operator will receive no calls in that time interval tomorrow.
 - In the next three days, the operator will receive a total of 1 call in that time interval.

Ans: (i) 0.0497 (ii) 0.0011